DSA Day 30: B+ Trees & Digital Search Trees

Book: Data Structures Through C in Depth - Ch6 §6.22-6.23 · pp 318-325 | Code in Python

Overview

A B+ tree stores all data in linked leaves (fast range scans). A trie/digital tree branches on symbols of the key.

Key concepts to master

- Data in leaves, linked leaves
- Range queries
- Trie insert/search

Python implementation

```
class Trie:
    def __init__(self): self.ch={}; self.end=False
    def insert(self,w):
        n=self
        for c in w: n=n.ch.setdefault(c,Trie())
        n.end=True
```

Complexity

Trie ops O(len(word)); B+ search O(log n).

Common pitfalls

- Confusing B-tree with B+ tree
- Forgetting the end-of-word flag in a trie

Practice

- Range-search a B+ tree
- Insert words into a trie